
Algorithmic Fairness in Credit Scoring: Does Model Compression Amplify Bias?

D1 Report - Dataset Analysis, EDA, and Preliminary Implementation

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Abstract

This report presents the D1 findings for Project P26: Algorithmic Fairness in Credit Scoring. We investigate whether compressing a Multi-Layer Perceptron (MLP) using pruning and quantization amplifies existing demographic bias in credit-scoring models. Using two publicly available datasets - the UCI German Credit dataset and the UCI Adult dataset - we perform exploratory data analysis (EDA), implement a Random Forest baseline and an MLP with two hidden layers, and report preliminary results across five independent runs. Our EDA reveals a 7.5% gender bias gap in the German Credit dataset and a 19.5% gender bias gap in the UCI Adult dataset before any model is trained. The Random Forest achieves 78.3% accuracy on German Credit while the MLP achieves 71.4%, consistent with expectations for a small 1,000-sample dataset. On UCI Adult, the MLP (85.4%) outperforms the Random Forest (80.5%), demonstrating that neural networks benefit from larger datasets. These findings establish our fairness baselines for D2 compression experiments.

1 Introduction

Credit scoring systems determine access to loans, mortgages, and financial services for millions of people. These systems increasingly rely on machine learning (ML) models that learn from historical data. A critical concern is whether these models encode and perpetuate existing societal biases - for example, by denying loans to women or minority groups at higher rates than equally qualified majority applicants.

This project addresses a specific and underexplored question: *does compressing a credit-scoring MLP (through pruning or quantization) make it more or less fair toward demographic groups?* Prior work by Hooker et al. [3] demonstrated that compression amplifies bias in image recognition models, but this has not been studied specifically for credit scoring with explicit demographic fairness attributes.

Our research questions are:

1. What is the baseline fairness gap in raw credit scoring datasets?
2. How do classical ML models compare to MLPs on accuracy and fairness?
3. (D2) Does MLP compression increase or decrease these fairness gaps?

The video submission is available at: https://drive.google.com/file/d/1w14-j55qmSXWjVD9r3JJNd6_CF8LGdL0/view?usp=sharing

2 Related Work

Hardt et al. (NeurIPS 2016) - Equality of Opportunity in Supervised Learning [1]. This foundational paper defined *equalized odds* and *equal opportunity* as formal fairness metrics for ML classifiers. They showed that demographic parity violations exist in real-world datasets including credit data. We adopt their equalized odds definition to measure fairness in our experiments. Dataset used: UCI Adult (our Dataset 2).

Agarwal et al. (ICML 2019) - A Reductions Approach to Fair Classification [2]. Agarwal et al. proposed a practical method to train fair classifiers by reducing the fairness problem to a series of standard weighted classification problems. They tested on UCI Adult - our Dataset 2 - and reported accuracy-fairness tradeoff curves that serve as our baseline comparison.

Hooker et al. (ICLR 2020) - Characterising Bias in Compressed Models [3]. This is the most directly relevant paper to our project. Hooker et al. showed that pruning and quantization disproportionately degrade performance for underrepresented groups, effectively amplifying existing bias. They tested on image datasets (CelebA). Our project extends this finding to credit scoring tabular data with explicit age and gender fairness attributes - a domain they did not study.

Donini et al. (NeurIPS 2018) - Empirical Risk Minimization Under Fairness Constraints [4]. Donini et al. trained fairness-constrained SVMs on the UCI German Credit dataset - our Dataset 1 - reporting approximately 0.73 accuracy. Their results provide a direct numerical comparison for our SVM experiments in D2.

Woodworth et al. (NeurIPS 2017) - Learning Non-Discriminatory Predictors [5]. This paper studied the theoretical tradeoff between accuracy and fairness on UCI Adult, showing that higher accuracy models tend to encode more bias on imbalanced data. Their framework guides our interpretation of the accuracy-fairness tradeoff in D2.

Zafar et al. (AISTATS 2017) - Fairness Constraints: Mechanisms for Fair Classification [6]. Zafar et al. tested fairness-constrained logistic regression and SVM on the German Credit dataset and reported accuracy around 0.72–0.75. These numbers provide a second direct comparison for our D2 SVM and Logistic Regression results.

3 Datasets

3.1 Dataset 1 - UCI German Credit Dataset

The German Credit dataset [7] contains records of 1,000 bank customers in Germany. Each record includes 20 features describing financial and personal attributes, and the binary label indicates whether the customer is a good (0) or bad (1) credit risk.

Key statistics:

- Samples: 1,000 | Features: 20 | Task: Binary classification
- Missing values: None - completely clean
- Class balance: 700 good credit (70%) vs 300 bad credit (30%) - imbalanced

Fairness attributes: Age (binarized: Young <35 vs Senior ≥ 35) and Gender (extracted from the personal status/sex feature: 310 Female, 690 Male).

Fairness finding from EDA: Female applicants have a bad credit rate of 35.2% compared to 27.7% for male applicants - a **7.5% gender bias gap** already present in the raw data before any model is trained.

3.2 Dataset 2 - UCI Adult (Income) Dataset

The UCI Adult dataset [7] contains US Census data for 48,842 adults. The task is to predict whether a person earns more than \$50,000 per year. It contains explicit gender (sex) and race attributes, making it the most widely used benchmark in fairness research [1, 2].

Key statistics:

- Samples: 48,842 | Features: 14 input features | Task: Binary classification
- Missing values: Present in workclass, occupation, native_country - fixed using mode imputation
- Class balance: $\leq 50K = 76.07\%$ vs $> 50K = 23.93\%$ - imbalanced

Fairness attributes: Sex (Male/Female) and Race (binarized: White vs Non-White).

Fairness findings from EDA:

- Gender gap: Males earn $> \$50K$ at 30.4% vs Females at 10.9% - **19.5% gender bias gap**
- Race gap: White at 25.4% vs Non-White at 15.3% - **10.1% race bias gap**

4 Methodology

4.1 Experimental Setup

All experiments use five independent runs with fixed random seeds $\{42, 0, 1, 2, 3\}$ for reproducibility. We report mean $\pm$ standard deviation across all five runs. Data is split 80% training / 20% test with stratification to preserve class balance. All features are standardized using `StandardScaler`. Categorical variables are encoded using `LabelEncoder`.

4.2 Classical ML - Random Forest

We train a Random Forest classifier [8] with 100 trees, maximum depth of 10, and balanced class weights to handle the class imbalance present in both datasets. This serves as the classical ML baseline for D1.

4.3 MLP Architecture

Both datasets use the identical MLP architecture:

$$\text{Input} \rightarrow \text{Linear}(d, 64) \rightarrow \text{ReLU} \rightarrow \text{Dropout}(0.3) \rightarrow \text{Linear}(64, 32) \rightarrow \text{ReLU} \rightarrow \text{Dropout}(0.3) \rightarrow \text{Linear}(32, 1) \quad (1)$$

where d is the input dimensionality. We use the Adam optimizer [9] with learning rate 10^{-3} and weight decay 10^{-4} . The loss function is `BCEWithLogitsLoss` with `pos_weight` to handle class imbalance.

Epoch strategy: German Credit uses **fixed 50 epochs** (small dataset, 1,000 samples, model converges by epoch 40). UCI Adult uses **early stopping** with maximum 200 epochs and patience of 10 (large dataset, 48,842 samples, unknown convergence point).

4.4 Evaluation Metrics

We report three metrics for all models:

- **Accuracy** - overall correctness of predictions
- **F1 Score** - harmonic mean of precision and recall, robust to class imbalance
- **AUC-ROC** - area under the receiver operating characteristic curve

5 Experiments and Results

5.1 German Credit Dataset Results

Table 1 reports preliminary results on the German Credit dataset.

Table 1: D1 Results - German Credit Dataset (mean $\pm$ std, 5 seeds, fixed 50 epochs)

Model	Accuracy	F1 Score	AUC-ROC
Random Forest	0.7830 ± 0.0180	0.6001 ± 0.0470	0.8009 ± 0.0270
MLP (2 hidden layers)	0.7140 ± 0.0320	0.5892 ± 0.0530	0.7885 ± 0.0230
Donini et al. [4] SVM	~ 0.730	-	-
Zafar et al. [6] LR/SVM	$\sim 0.720\text{--}0.750$	-	-

The Random Forest outperforms the MLP on all three metrics. This is expected - neural networks typically require more than 1,000 samples to outperform classical methods on tabular data. Our Random Forest baseline (0.783) exceeds the baselines reported by Donini et al. [4] (~ 0.730) and Zafar et al. [6] ($\sim 0.72\text{--}0.75$) on the same dataset, noting that those papers applied fairness constraints which reduce accuracy.

5.2 UCI Adult Dataset Results

Table 2 reports preliminary results on the UCI Adult dataset.

Table 2: D1 Results - UCI Adult Dataset (mean $\pm$ std, 5 seeds, early stopping)

Model	Accuracy	F1 Score	AUC-ROC
Random Forest	0.8054 ± 0.0012	0.6822 ± 0.0021	0.9171 ± 0.0010
MLP (2 hidden layers)	0.8540 ± 0.0018	0.6595 ± 0.0030	0.9098 ± 0.0015

On UCI Adult, the MLP outperforms the Random Forest on Accuracy (0.854 vs 0.805), demonstrating that the larger dataset (48,842 samples) provides enough data for the neural network to learn effectively.

5.3 Fairness Analysis - Raw Data

Table 3 summarises the demographic bias gaps found in the raw data during EDA.

Table 3: Fairness bias gaps found in raw data (EDA findings)

Dataset	Attribute	Group 1	Group 2
German Credit	Gender (bad credit rate)	Female: 35.2%	Male: 27.7%
	Gap	7.5% - females labeled bad more often	
UCI Adult	Gender (>\$50K income rate)	Male: 30.4%	Female: 10.9%
	Gap	19.5% - females earn less often	
	Race (>\$50K income rate)	White: 25.4%	Non-White: 15.3%
	Gap	10.1% - Non-White earn less often	

These findings confirm the demographic disparities formally defined by Hardt et al. [1]. The 7.5% gender gap in German Credit directly demonstrates the type of disparity their equalized odds metric is designed to detect.

6 Discussion

Why Random Forest outperforms MLP on German Credit but not on UCI Adult: The German Credit dataset has only 1,000 samples. With such limited data, the MLP (0.714) underperforms

the Random Forest (0.783). On the larger UCI Adult dataset (48,842 samples), the MLP (0.854) outperforms the Random Forest (0.805), confirming that dataset size is a key determinant of MLP performance on tabular data. This is consistent with Woodworth et al. [5], who noted that model capacity must be matched to dataset size.

Bias is present before training: Both datasets show measurable demographic bias in the raw data. The 7.5% gender gap in German Credit and 19.5% gender gap in UCI Adult exist independently of any model. Any bias amplification caused by compression (D2) will be measured relative to these baselines.

Limitations: This D1 report uses only one classical ML method (Random Forest) and one MLP without compression. D2 will extend to SVM, KNN, Logistic Regression, and Decision Tree, and will apply pruning, quantization, and coreset selection to the MLP.

7 Conclusion

We have established the D1 baselines for Project P26. Both datasets contain measurable demographic bias before any model is trained - 7.5% and 19.5% gender gaps respectively. The Random Forest outperforms the MLP on the small German Credit dataset, while the MLP wins on the larger UCI Adult dataset. In D2, we will compress the MLP using pruning and quantization and measure whether these bias gaps increase or decrease - directly extending the findings of Hooker et al. [3] to the credit scoring domain.

References

- [1] Hardt, M., Price, E., & Srebro, N. (2016). Equality of opportunity in supervised learning. *Advances in Neural Information Processing Systems (NeurIPS)*, 29.
- [2] Agarwal, A., Beygelzimer, A., Dudik, M., Langford, J., & Wallach, H. (2019). A reductions approach to fair classification. *International Conference on Machine Learning (ICML)*.
- [3] Hooker, S., Moorosi, N., Clark, G., Bengio, S., & Learned-Miller, E. (2020). Characterising bias in compressed models. *arXiv preprint arXiv:2010.03058*.
- [4] Donini, M., Oneto, L., Ben-David, S., Shawe-Taylor, J., & Pontil, M. (2018). Empirical risk minimization under fairness constraints. *Advances in Neural Information Processing Systems (NeurIPS)*, 31.
- [5] Woodworth, B., Gunasekar, S., Ohannessian, M. I., & Srebro, N. (2017). Learning non-discriminatory predictors. *NeurIPS Workshop*.
- [6] Zafar, M. B., Valera, I., Rodriguez, M. G., & Gummadi, K. P. (2017). Fairness constraints: Mechanisms for fair classification. *AISTATS*.
- [7] Lichman, M. (2013). UCI Machine Learning Repository. <https://archive.ics.uci.edu>.
- [8] Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32.
- [9] Kingma, D. P., & Ba, J. (2014). Adam: A method for stochastic optimization. *arXiv:1412.6980*.
- [10] Kamiran, F., & Calders, T. (2012). Data preprocessing techniques for classification without discrimination. *Knowledge and Information Systems*, 33(1), 1–33.